WhatsApp Order Management System: Project Plan

Version 1, prepared 8 November 2026. Covers RP design orders from WhatsApp, on-site orders, designer allotment, output tracking and history.

1. Purpose and scope

Today only the owner's WhatsApp receives customer orders. This system records every incoming WhatsApp message automatically, turns it into an order the owner can confirm in a few taps, lets the owner or staff allot the work to a designer manually, collects the designer's finished images and videos, and sends the approved result back to the customer. It also keeps a searchable history of all orders.

In scope

- WhatsApp intake (one number now, more numbers later)
- On-site orders entered manually
- Order details (sizes, weight, quantity, notes) shown to the designer together with the customer's own messages
- Manual allotment to 10-12 designers, with a compact view of each designer's work today
- Designer output upload (any number of images and videos), owner approval, delivery to the customer
- History per day and per month, and search by designer, date range, category, customer or image
- Admin, owner/staff and designer panels, English with an Urdu button

Out of scope for now

- Pricing and payments
- Deadlines (the owner chose not to track them)
- Automatic parsing of catalog orders (catalog orders are entered as plain fields: book name and item number)
- Importing old data (there is none)

2. Roles and permissions

Role	Who	Can do
Admin	Developer	Views and monitors all data with the owner's permission; technical settings (WhatsApp connection, AI key, logs); emergency account override
Owner	Business owner	Everything business-related, including creating, editing and deleting designer and staff accounts
Staff	Staff member covering the owner	Everything the owner can do except creating, editing or deleting accounts
Designer	10-12 designers	See only the orders allotted to them; upload outputs; mark work submitted

Notes:

- Owner and staff each get their own login, so the history shows who did what.
- Designers never see the customer's phone number.
- The owner has a setting to switch off admin access to data. Every admin view of order data is written to an access log. This protects both the owner and the admin.
- Assumption to confirm: the owner creates staff accounts.

3. Order lifecycle

Orders are split into items. Each image is an item with its own category, designer, details and status.

Item statuses: New, Allotted, In progress, Submitted, Approved, Delivered, plus Change requested.

1. **New.** Created automatically in the Inbox from incoming messages.
2. **Allotted.** The owner or staff picks a designer manually. A busy designer never blocks assignment; the screen only shows each designer's open count.
3. **In progress.** The designer starts work.
4. **Submitted.** The designer uploads any number of images and videos and submits.

5. **Approved.** The owner or staff approves the output. Alternatively they set Change requested, which returns the item to the designer with a note.
6. **Delivered.** The system sends the approved media to the customer on WhatsApp and records it.

Grouping rule. Messages from the same customer arriving within a short quiet period (default 3 minutes, admin-adjustable) form one draft order. Each image becomes an item. Voice notes and text are attached to the order. If a voice note replies to a specific image, WhatsApp reports which message it replies to, so the note attaches to that item.

Follow-up rule (adjustable).

- A text or voice message from a customer with an open order sets Change requested and shows a badge with the new messages. No new order is created.
- A new image from that customer goes to the Inbox as a new draft, with a one-tap merge into the open order.
- After an order is Delivered, new messages start a new draft. The owner can link it to the earlier order as a repeat or revision.

Acknowledgement. When a draft is created, the customer receives one short automatic Urdu reply with the order number, for example: آپ کا آرڈر موصول ہو گیا ہے۔ آرڈر نمبر 123۔ شکریہ۔ It is sent once per group of messages, not once per message.

4. Order details shown to the designer

Customers give details randomly, so every detail field is optional. The designer's order screen shows, from top to bottom:

1. The detail fields that were filled in, in large text. Empty fields are hidden.
2. Images and videos.
3. The customer's messages and voice notes in time order, with play buttons.

Category-based fields. Each category has its own field list, editable by the owner or admin without code changes. Definitions live in a `category_fields` table; values live in a JSONB `specs` column on each item.

Fields every category can use: weight, quantity, pair (yes/no), stone size, free-text note.

Starting suggestions per category (to be reviewed with the owner):

Category	Extra fields
Ring	Ring size standard (steel or plastic), size number, thawa (top part) size in mm or inch
Bangle	Size in inches and points, entered as the customer says it
Pendant, necklace, chain	Chain length (chains are usually bought from wholesalers, so this is mainly informational)
Earrings	Pair, stone size

Units. Weight accepts gram, tola, masha and ratti, because the owner uses whatever the customer uses. Sizes accept mm or inch where relevant. The system stores the value and the unit exactly as entered and never converts automatically in version 1. A silent conversion error would mean a wrong piece.

Who fills the fields. The owner or staff fills them when confirming the order, since they listen to the voice note to pick the category anyway. The designer can also play the voice note. In phase 2, AI may suggest values from voice notes, but suggested values are shown in a different color and need a human tap to confirm, because sizes are numbers that are easy to mishear.

Changes. When a follow-up changes a detail, the designer sees the old value crossed out beside the new one, a Change requested badge, and a Seen button.

Categories. An admin-managed list. The dropdown is searchable, shows the most-used categories first, and allows adding a new one on the spot.

5. WhatsApp integration

Official route. Use the WhatsApp Business Platform (Cloud API). Unofficial libraries that imitate WhatsApp Web violate the terms of service and risk a ban on the company's main sales channel, so they are not recommended.

Keeping the owner's number. Meta's coexistence feature lets a number that is already on the WhatsApp Business app connect to the Cloud API while staff keep using the app. Requirements reported by providers:

- The number must be on the WhatsApp Business app, not regular WhatsApp.
- The app must be version 2.24.17 or newer.

- The app should be opened at least once every 14 days to keep the link active.

Open question for the owner: is the number on WhatsApp Business or regular WhatsApp? If it is regular WhatsApp, a new number or a migration is needed.

Customer service window and cost. A customer's message opens a 24-hour window in which free-form replies are allowed, and each new customer message resets it. From 1 October 2026, providers report that free-form replies are free only up to 1,000 per number per month and billed per message after that, at rates that depend on the customer's country. Template messages are charged per delivered message. Check Pakistan's current rates on Meta's pricing page before budgeting.

Estimate: 16-20 orders a day, each with one acknowledgement plus the output images or videos, is roughly 1,500-3,000 outbound messages a month. This is above the free allowance, but the extra cost is expected to be small.

Delivery rule. Because most orders are delivered the next day, the window is often still open. The system sends free-form media if the window is open. If it is closed, the system shows the owner a warning and sends a pre-approved template message (for example, "Your design is ready"), then sends the media when the customer replies. Templates must be approved by Meta in advance.

Library. PyWa is an open-source (MIT) Python framework for the Cloud API that plugs into an existing FastAPI app and supports async. Use it for sending, receiving and media handling.

Multiple numbers. Every message and order stores a `whatsapp_account_id` from day one. Adding a second number later is then a settings change.

6. Screens

Owner and staff

- Inbox: new drafts with images, voice players and the customer's messages; confirm category and details; merge or split items.
- Today board: all of today's items by status (New, Allotted, In progress, Submitted, Approved, Delivered, Change requested).
- Allot screen: pick a designer; each designer's open count is shown as information only; "assign all to one designer" button.
- Designers today: compact cards, one per designer, listing the items they are working on now.

- Order detail: details, media, message thread, status history, approve, request change, send to customer.
- Add order: manual entry for on-site customers, with the same detail fields and image upload.
- History: per day and per month, with filters.
- Customers: phone, name, and past orders, so repeat orders are easy to find.
- Designer accounts (owner only).

Designer

- My work: allotted items, newest changes highlighted.
- Item screen: details, reference images, customer messages and voice notes, upload output (images and videos), submit.

Admin

- Overview and monitoring, categories and fields, WhatsApp settings, AI settings, logs and weekly report, access log.

All panels are responsive: cards on phones, tables and multi-column layouts on PC.

Urdu button. The interface uses translation files (react-i18next) with right-to-left layout and an Urdu font. Google's page translate widget is not used, because it garbles jewelry terms and breaks layouts. Only the interface is translated, not customer messages.

7. Search and history

- Filters: designer, date range (for example February to April), category, customer, order number, status.
- Counts per day, per month, per designer and per category.
- Image search (phase 3): each image gets a CLIP embedding stored in PostgreSQL with the pgvector extension. Uploading an image returns the nearest matches. The same lookup warns when a new design resembles an earlier order, for example "this customer ordered something similar before".

8. Data model (main tables)

- `whatsapp_accounts`: id, name, phone_number_id, business_account_id, active
- `customers`: id, phone (unique), profile_name, display_name, notes

- `messages`: id, account_id, customer_id, wa_message_id (unique, prevents duplicate webhooks), direction, type, text, media_id, reply_to_wa_id, timestamp, raw JSONB, order_id, item_id
- `orders`: id, number, account_id, customer_id, source (whatsapp or onsite), created_by, created_at, delivered_at, deleted_at
- `order_items`: id, order_id, category_id, designer_id, status, specs JSONB, note, change_flag, created_at
- `media`: id, storage_key, mime, size, sha256, kind (customer or output), uploaded_by, created_at
- `item_media`: item_id, media_id, role (reference or output)
- `categories` and `category_fields`: category list and per-category field definitions
- `users`: id, username, password_hash, role, full_name, active, last_login
- `status_history`: item_id, from_status, to_status, user_id, at
- `audit_log` and `error_log`: who did what, and every failure
- `ai_prompt_versions` and `ai_corrections`: versioned instructions and human corrections of AI suggestions (phase 2)
- `embeddings`: media_id, vector (phase 3)

Orders are never hard-deleted; a `deleted_at` flag keeps history intact.

9. Architecture and stack

- **Backend:** FastAPI, PostgreSQL with pgvector, SQLAlchemy 2 and Alembic migrations, Pydantic.
- **Worker:** a background worker (ARQ with Redis is a small option). The webhook must save the raw message and return 200 immediately, because Meta retries slow webhooks, and background tasks inside the web process are lost on restart.
- **Media:** download WhatsApp media immediately (links are short-lived), store in S3-compatible object storage, and convert voice notes to a format every browser can play (WhatsApp voice notes are Ogg Opus, which some browsers cannot play) using ffmpeg. Compress videos before sending to WhatsApp, which limits media size; keep the originals.
- **Frontend:** React with Vite and TypeScript, Tailwind CSS, TanStack Query, react-i18next. Live updates through polling every 30 seconds at first.

- **Deployment:** one small VPS with Docker Compose, Nginx and HTTPS. At 16-20 orders a day, microservices and Kubernetes are not needed.

10. Security and privacy

- Passwords hashed with argon2 or bcrypt; login rate limiting; HTTPS everywhere.
- Role checks on every API endpoint, not only in the interface. Hiding a button is not security.
- Verify the signature on every WhatsApp webhook.
- Secrets (WhatsApp token, AI key) only in environment variables or encrypted settings, never shown again after saving.
- Media served through short-lived signed links.
- Designers cannot see customer phone numbers.
- Audit log of status changes, assignments, account changes and admin data views.

11. Logging and the improvement loop

The system should improve, but only with human approval. Letting an AI change its own instructions from customer messages is unsafe: those messages are untrusted text, a bad change could quietly make orders worse, and costs could grow unwatched.

1. **Record everything from day one:** every message, every AI suggestion, every human correction (for example, "AI said ring, owner changed to bracelet"), and every error (failed send, failed upload, webhook error).
2. **Weekly report page for the admin:** error counts, how often suggestions were corrected, failed sends, items stuck in Change requested. No AI needed.
3. **AI settings page (admin only):** add an API key, choose a model, and edit the instruction box. Every save creates a new version. A "test on the last 20 real messages" button runs before activation, and rollback is one click.
4. **Optional "Suggest improvements" button:** sends the week's corrections to an LLM, which proposes new instruction text. The admin reviews the difference and approves it manually.

12. Hosting, backups and costs

Recurring costs to discuss with the owner: the VPS, WhatsApp message fees, object storage, and an optional AI API.

Storage planning: at an assumed 30 MB of media per order, growth is about 600 MB a day or 18 GB a month. Use object storage or a large disk.

Backups: a daily database dump and media copy to a second location. The records are the business history, so a failed VPS must not lose them.

Downtime: under coexistence the owner's phone still receives messages if the server is down, and Meta retries failed webhook deliveries for a limited time (check the exact window in Meta's documentation).

13. Phases and tasks

Phase 0: before coding

- Confirm with the owner whether the number is on WhatsApp Business or regular WhatsApp.
- Agree hosting and monthly budget.
- Create the Meta business account and begin verification (this can take time).
- Collect 30-50 real voice notes, with permission, and note what they usually say.
- Collect the full category list and the exact size phrases used for rings and bangles.

Phase 1: core, no AI

1. Project setup, database, migrations, login and roles.
2. WhatsApp webhook: signature check, save raw messages, ignore duplicates.
3. Media download and storage; voice conversion.
4. Grouping into draft orders and the acknowledgement reply.
5. Inbox: confirm category and details.
6. Category and field management.
7. Allot screen, Today board, Designers today.
8. Designer panel with upload and submit.
9. Approve, request change, send to customer (free-form, template fallback).
10. Manual on-site order entry.
11. Customers page and history filters.

12. English and Urdu interface; phone and PC layouts.
13. Logging, error log, audit log, backups.

Phase 2: assistance

- Category suggestion from the image.
- Optional voice transcription and detail suggestions, tested on real voice notes first, always human-confirmed.
- Weekly report page and AI settings with prompt versions.

Phase 3: growth

- Image similarity search and repeat-order warnings.
- Monthly reports per designer and per category.
- Second WhatsApp number.

14. Risks

- **Number type or Meta verification delays.** Mitigation: start Phase 0 first.
- **Punjabi and Urdu voice recognition is unreliable.** Mitigation: voice is stored and played, never required to be transcribed.
- **Wrong grouping of messages.** Mitigation: adjustable quiet period, merge and split buttons.
- **Storage growth.** Mitigation: object storage and monitoring.
- **Single server failure.** Mitigation: daily backups and a documented restore.
- **Staff adoption.** Mitigation: a short confirm screen with large buttons and minimal required fields.

15. Open questions

- The exact phrases customers use for ring and bangle sizes, including the term "cheen sitar".
- Whether the owner creates staff accounts.
- Whether the follow-up rule for new images (new draft with merge) suits the owner.
- Whether the template message should be in Urdu script or Roman Urdu.
- The final category list.
- Hosting provider and monthly budget.

16. References

- WhatsApp Business Platform pricing: https://developers.facebook.com/documentation/business-messaging/whatsapp/pricing
- PyWa (Python framework for the Cloud API): https://github.com/david-lev/pywa
- Coexistence overview from a messaging provider: https://developer.8x8.com/connect/docs/whatsapp/whatsapp-business-app-coexistence
- Pricing change summary from a messaging provider: https://m.aisensy.com/blog/whatsapp-business-pricing/